MANIPAL INSTITUTE OF TECHNOLOGY

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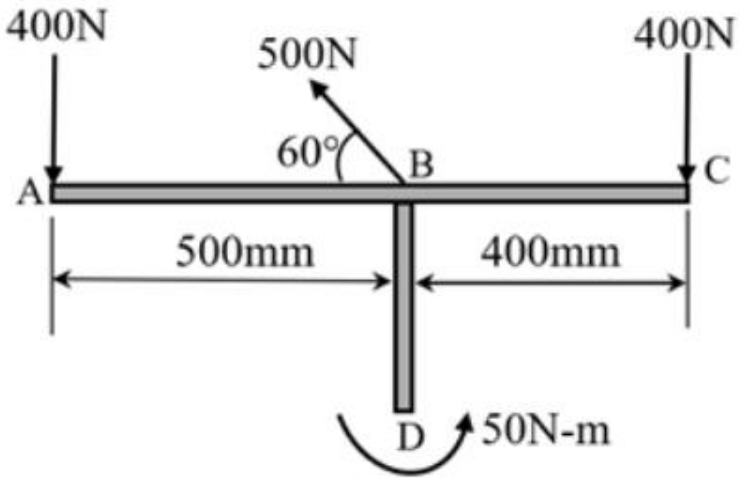
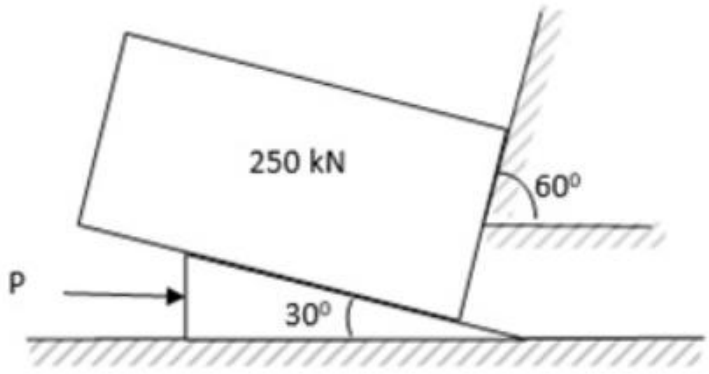
DEPARTMENT OF CIVIL ENGINEERING

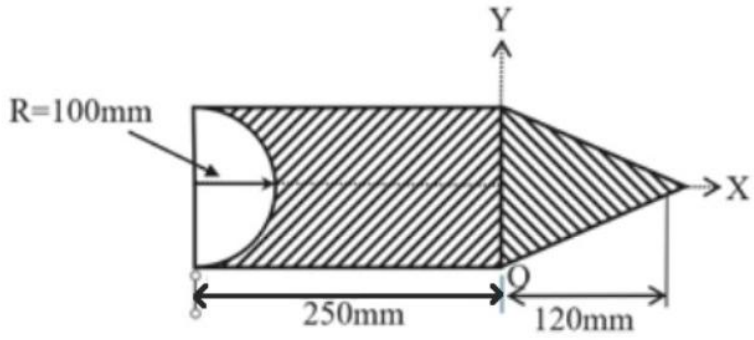
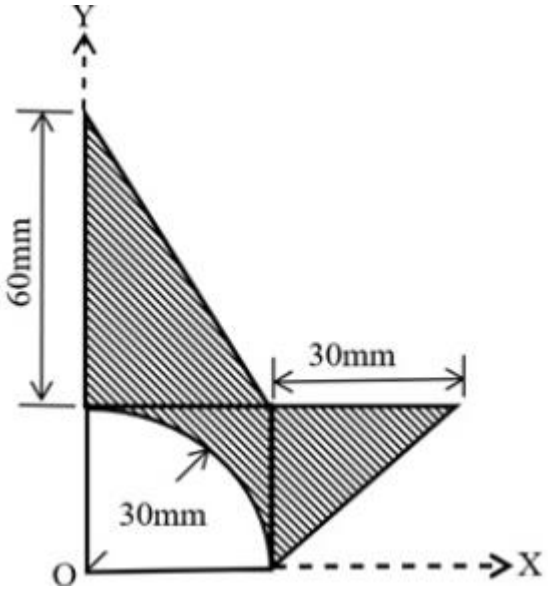
I Semester B.Tech. (CIVIL ENGINEERING)

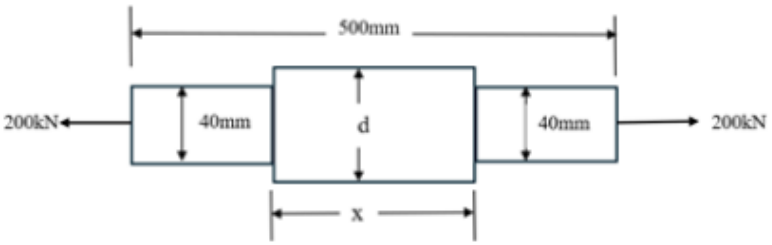
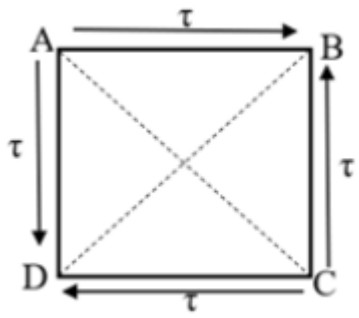
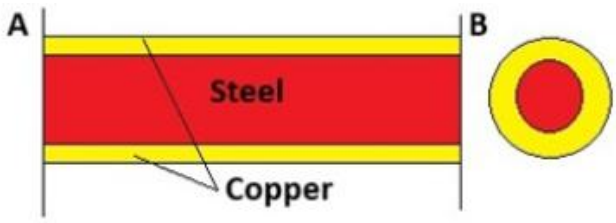
END SEM EXAM

Course: Mechanics of Solids **Course Code: CIE-1071**

Time: 09:30AM **Date: 28/11/2024** **Max. Marks: 50**

Q. No.	Question	Marks	NBA		IET*		BL
			CO	PO	CLO	AHEP4 LO	
1A	A bracket is subjected to a coplanar force system as shown in the figure. Determine the magnitudeand the position of resultant w.r.t point B. Locate the resultant in the figure. 	5	1	PO1, PO2, PO12, PSO2	1	1,2, 18	2
1B	Determine the force required to impend the block of 250 kN up the plane by driving a wedge as shown in the figure. Ignore the weight of the wedge and assume a coefficient of friction of 0.25 for all contact surfaces. 	5	2	PO1, PO2, PO12, PSO2	2	1,2, 18	3

2A	Compute centroid of the shaded area in figure with respect to O-Y axis shown. 	**3**	3	**PO1, PO2, PO12, PSO2**	**3**	**1,2, 18**	**2**
2B	Compute the moment of inertia of the shaded area shown in figure about the OX axis shown. 	**5**	3	**PO1, PO2, PO12 , PSO2**	**3**	**1,2, 18**	**3**
2C	Define centroid and write the significance of axis of symmetry in locating centroid of any geometrical shape.	**2**	3	**PO1, PO2, PO12 , PSO2**	**3**	**1,2, 18**	**2**
3A	A cylindrical boiler is 600mm in diameter and 1m length. It is required to withstand a pressure of 120m of water. If the permissible tensile stress is 25N/mm2, permissible shear stress is 10N/mm2 and permissible change in diameter is 0.2mm, calculate the minimum thickness of the metal required. Take E = 100GPa, and $\mu = 0.3$.	**5**	4	**PO1, PO2, PO12 , PSO2**	**4**	**1,2, 18**	**3**
3B	Determine an expression for the total extension of the tapered bar of length L, with rectangular cross section, having uniform thickness 'b' as shown in figure, when subjected to an axial tensile load W.	**5**	4	**PO1, PO2, PO12 , PSO2**	**4**	**1,2, 18**	**2**

4A	For the bar shown below, determine the diameter of the central portion and its length, if the total extension of the bar is 0.20 mm. Take E=200 GPa. Stress at central portion is limited to 70 N/mm2 	**4**	4	**PO1, PO2, PO12 , PSO2**	**4**	**1,2, 18**	**3**
4B	Illustrate with suitable derivation that "The intensity of normal tensile stress developed on plane BD is numerically equal to the intensity of shear stress". 	**3**	4	**PO1, PO2, PO12 , PSO2**	**4**	**1,2, 18**	**3**
4C	Using the available relation between E, G, μ, and E, K, μ determine the relation between E, G, K. Where E is the modulus of elasticity, G is the modulus of rigidity, K is bulk modulus and μ is the Poisson's ratio.	**3**	4	**PO1, PO2, PO12 , PSO2**	**4**	**1,2, 18**	**2**
5A	Steel bolt of diameter 12 mm pass through a brass tube of internal diameter 18 mm and external diameter 30 mm. The bolt is tightened by a nut so that the length of tube is reduced by 2 mm. compute the axial stress in the bolt and the tube. If the initial length of the tube is 1.8m, Es= 200GPa and Ebr=120GPa	**5**	5	**PO1, PO2, PO12 , PSO2**	**5**	**1,2, 18**	**3**
5B	A circular steel bar is inserted into the copper tube and fixed between the two supports as shown in the figure. Steel bar has a diameter of 60 mm, thickness of the copper tube is 40 mm. If the temperature is raised by 70oC, determine the stress in each metal. The length of the bar at normal temperature is 1.2m. E s = 200GPa, E cu =100GPa, α s =12 x 10-6/°C, α cu =17 x 10-6/°C 	**5**	5	**PO1, PO2, PO12 , PSO2**	**5**	**1,2, 18**	**3**